TADALAFIL

No nitrates for 48 h

Lasts 36 h

CHEMICAL NAME (6R,12aR)-6-(1,3-benzodioxol-5-yl)-2-methyl-2,3,6,7,12,12a-hexahydropyrazino[1′,2′:1,6]pyrido[3,4-b]indole-1,4-dione

CLASS Long-acting phosphodiesterase type 5 (PDE-5) inhibitor

TRADE NAMES Cialis · Adcirca (PAH) · Tadacip

THERAPEUTIC USES / INDICATIONS

- Erectile dysfunction (on demand or daily).
- Lower urinary tract symptoms of BPH (daily low dose).
- Pulmonary arterial hypertension.

PHARMACODYNAMICS

- Inhibits **PDE-5** → ↑ cGMP → smooth-muscle relaxation in penis, prostate, bladder and pulmonary vessels.

ADMINISTRATION

- **PO** with or without food (not affected by food).
- On demand ≥30 min before activity, or daily at the same time.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, dyspepsia.
- Back pain, myalgia (PDE-11).
- Flushing, nasal congestion.

ADVERSE EFFECTS (SERIOUS)

- Severe hypotension with nitrates.
- Priapism.
- NAION; sudden hearing loss.

DOSAGES

- **ED on demand:** 10 mg (up to 20 mg) at least 30 min before; max once daily.
- **ED daily / BPH:** 2.5–5 mg once daily.
- **PAH:** 40 mg once daily.
- Renal/hepatic impairment: lower doses.

CONTRAINDICATIONS & CAUTIONS

- Nitrates (within 48 h of tadalafil); riociguat.
- Severe cardiac disease; recent MI/stroke; hypotension.
- Previous NAION.

KEY INTERACTIONS

- **Nitrates:** contraindicated.
- **α-blockers:** hypotension.
- **CYP3A4 inhibitors/inducers:** dose adjustments.

PHARMACOKINETICS

- **Onset:** 30 min. **Duration:** up to 36 h.
- Metabolised by CYP3A4.
- Excreted in faeces. **t½:** 17.5 h.

NURSING MANAGEMENT

- Document PDE-5 use; nitrates must not be given for 48 h.
- Monitor BP; assess urinary symptoms (BPH).

PATIENT EDUCATION

- Effect lasts up to 36 hours.
- Do not take nitrate medicines for 48 hours after a dose; tell emergency staff.
- Seek urgent help for an erection >4 h or sudden vision loss.

KEY POINTS

- Long-acting PDE-5 inhibitor (36 h).
- No nitrates for 48 h.
- ED, BPH (5 mg daily), PAH.
- Back pain/myalgia.

SOURCES CHECKED **[710]** Tadalafil 10 mg film-coated tablets (UK SmPC) **[711]** tadalafil, oral (US label) **[712]** Tadalafil, oral (US label)

Full references p. 432–457